

“Buniyadi”

A Unified Startup Intelligence Platform

Software Requirements Specification

for Business Ideation, Market Validation, Strategic Positioning,
Brand Development, Competitor Analysis, and Business Documentation

Category: Branding and Marketing

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1. Introduction

1.1 Purpose

This Software Requirement Specification (SRS) document outlines the requirements for developing “Buniyadi”, a centralized, AI-powered startup workspace web application built using the MERN stack (MongoDB, Express.js, React.js, Node.js) and FastAPI. The primary goal of Buniyadi is to serve as the single source of truth for founders, transforming a core business idea into a complete, continuously evolving strategy, brand, and documentation ecosystem.

1.2 Scope

The scope of this project encompasses the design, development, and deployment of a web-based workspace catering to the following user groups:

- Founders: Who can input ideas conversationally, generate comprehensive startup assets, and manage their business evolution.
- Co-founders / Teams: Who can collaborate within the workspace.
- Advisors / Investors: Who can be granted read-only access to view living business documents and timelines.

This document covers all functional and non-functional requirements, constraints, high-level MVC design, and the Agile development plan.

1.3 Definitions, Acronyms, and Abbreviations

- MERN: MongoDB, Express.js, React.js, Node.js.
- FastAPI: A modern Python web framework used as a dedicated AI microservice.
- MVC: Model-View-Controller architectural pattern.
- SSOT: Single Source of Truth.
- LLM: Large Language Model.
- RAG: Retrieval-Augmented Generation — a technique to ground AI responses in stored knowledge.

1.4 References

- MERN Stack documentation: MongoDB, Express.js, React.js, Node.js.
- FastAPI documentation (fastapi.tiangolo.com).

1.5 Overview

Section 2 provides an overall description of the product. Section 3 details functional and non-functional requirements. Section 4 outlines the technology stack and architecture. Section 5 lists all 25 features. Section 6 presents the Agile development plan.

2. Overall Description

2.1 Product Perspective

Buniyadi is a standalone web application. The core application uses the MERN stack for user management, data storage, and the frontend interface. It connects to a FastAPI microservice that handles all AI-related tasks, including prompt engineering and LLM communication. No other frameworks or backend services are used.

2.2 Product Features

Buniyadi provides ten primary capabilities:

1. Conversational onboarding — ideas entered in plain text, no forms required.
2. Progressive context rendering — dynamic generation of business cards in real time.
3. Context-aware global memory — all prior decisions inform every future AI output.
4. Dynamic workspace rooms — dedicated spaces for Strategy, Brand, Competitors, Marketing, Financials, and Launch.
5. Global variable syncing (Ripple Effect) — editing one variable updates all dependent modules automatically.
6. Explainable AI — every AI-generated decision includes the reasoning behind it.
7. Decision timeline — version-control history of all pivots and edits.
8. Living business documents — auto-updating pitch decks, business plans, and investor reports.
9. AI-ready export — export full startup context as Markdown or JSON.
10. Role-based collaboration — co-founder editing and read-only investor links.

2.3 User Classes and Characteristics

11. Founders (Primary Users): Non-technical or technical entrepreneurs seeking guided business building. They require an intuitive, frictionless interface.
12. Co-founders / Team Members: Collaborate in real time within the same workspace.
13. Mentors / Investors: Require streamlined, read-only access to executive summaries and pitch decks.
14. Administrators: System overseers managing users and system health.

2.4 Operating Environment

- Web Application: Accessible on modern browsers (Chrome, Firefox, Edge, Safari).
- Frontend: React.js (responsive, mobile-compatible).
- Backend (Core): Node.js and Express.js deployed in a cloud environment.
- Backend (AI): Python and FastAPI as a dedicated microservice.
- Database: MongoDB cluster (cloud-managed).

2.5 Constraints

- AI generation must render progressively to avoid long loading screens.
- The application must follow the MVC design pattern strictly.
- Development is scoped to 5 Agile sprints.

- Technology is limited to MERN stack and FastAPI — no additional backend frameworks.

2.6 Assumptions and Dependencies

- Users have a stable internet connection.
- Third-party LLM APIs (e.g., OpenAI, Groq) are available and reliable.
- Founders are the primary initiators of all workspace content.

3. System Requirements

3.1 Functional Requirements

3.1.1 Workspace Rooms (Modules)

FR-1: The system shall provide a Business Strategy Room auto-generating Problem, Solution, Unique Value Proposition, and Business Model.

FR-2: The system shall provide a Competitor Intelligence Room discovering real-world competitors and generating a comparison matrix of strengths, weaknesses, and pricing.

FR-3: The system shall provide an Audience & Persona Room generating editable customer personas with pain points, demographics, and buying behaviours.

FR-4: The system shall provide a Brand Identity Room generating brand voice, typography recommendations, and colour palettes with strategic reasoning.

FR-5: The system shall provide a Marketing & Launch Room generating Go-To-Market strategies, acquisition channels, and launch checklists.

FR-6: The system shall provide a Financials & Pricing Room suggesting revenue models and pricing tiers based on detected audience and competitor benchmarks.

3.1.2 The Ripple Effect & Global Syncing

FR-7: The system shall allow users to inline-edit any generated variable within a workspace room.

FR-8: Upon a manual edit, the system shall automatically trigger an AI recalculation of all interconnected rooms via FastAPI to reflect the change.

FR-9: The system shall display an explanation tooltip or text block for every AI-generated decision.

FR-10: The system shall detect strategic contradictions (e.g., premium pricing for a low-income audience) and prompt the user to resolve them.

FR-11: The system shall allow users to highlight any text block and prompt the AI to rewrite or expand only that section while maintaining global context.

3.1.3 Version Control & Tracking

FR-12: The system shall record every manual edit and AI generation in a chronologically ordered Decision Timeline.

FR-13: The system shall allow users to revert the entire workspace or individual rooms to any previous state on the timeline.

FR-14: The system shall display a dynamic “Welcome Back” summary on login, showing changes since the last session and suggesting next steps.

3.1.4 Authentication & Workspace Creation

FR-15: The system shall allow users to register and securely log in using email and password.

FR-16: The system shall ingest the founder’s startup idea via a single plain-text input field.

FR-17: The FastAPI service shall extract core business parameters (industry, model, audience, risks) from the input and return them to the MERN backend for storage.

FR-18: The system shall create a named workspace for each submitted idea and store it in MongoDB.

3.1.5 Documents, Export & Collaboration

FR-19: The system shall auto-generate and keep synchronized standard business documents: Executive Summary, Pitch Deck, and Business Plan.

FR-20: The system shall allow one-click export of the startup context as an optimized Markdown or JSON file.

FR-21: The system shall allow one-click PDF export of investor reports or brand guidelines.

FR-22: The system shall generate a secure, shareable read-only URL for specific workspace rooms.

FR-23: The system shall support real-time collaborative editing for multiple co-founders via WebSocket.

FR-24: The system shall implement RBAC with Co-Founder (full edit), Mentor (comment only), and Investor (view only) roles.

FR-25: The system shall provide a global search bar to find decisions, competitor data, or document content across all rooms and past versions.

3.1.6 Core AI & Onboarding

FR-26: Conversational Idea Ingestion – single text input, no forms.

FR-27: Progressive Context Rendering – cards appear in real time as AI processes

FR-28: Context-Aware Global Memory — RAG-based knowledge graph remembers all decisions.

FR-29: Knowledge Base Upload — founders upload PDFs/docs to ground the AI.

3.2 Non-Functional Requirements

NFR-1: Progressive rendering must display initial AI output within 5 seconds of idea submission.

NFR-2: The architecture must support horizontal scaling for concurrent AI requests.

NFR-3: All client-server communication must occur over HTTPS.

NFR-4: User passwords and business data must be encrypted at rest.

NFR-5: The system must maintain 99.9% uptime, excluding scheduled maintenance.

3.3 External Interface Requirements

- **UI-1:** A responsive React.js web interface adapting to desktop, tablet, and mobile viewports.
- **CI-1:** RESTful APIs for communication between the React frontend, Express backend, and FastAPI microservice.
- **SI-1:** Integration with at least one external LLM provider (e.g., OpenAI or Groq) via the FastAPI service.

4. Technology Stack & Architectural Overview

4.1 Technology Stack Components

MongoDB

Document database storing user profiles, workspace configurations, AI-generated content, decision timelines, and exported documents.

Express.js

Node.js framework providing the core RESTful API, enforcing business logic, and routing requests between the React frontend and the FastAPI microservice.

React.js

Frontend library responsible for the responsive, dynamic workspace UI including room rendering, progressive card generation, and the decision timeline view.

Node.js

JavaScript runtime powering the primary backend server.

FastAPI (Python)

Dedicated AI microservice handling natural language processing, prompt engineering, RAG-based context retrieval, and all communication with external LLM providers. FastAPI is the sole interface between the application and any AI model.

4.2 High-Level Architecture (MVC)

Model

Mongoose schemas defining the structure of Users, Workspaces, Timeline Events, Room Data, and Exported Assets. Responsible for all data validation and database interactions.

View

React components presenting the workspace dashboard, individual rooms, the decision timeline, and document previews to the user. All user interactions generate API calls to the Controller.

Controller

Express route handlers receive requests from the View, apply business logic, and either write/read from MongoDB or forward AI processing requests to the FastAPI microservice. FastAPI endpoints are the only bridge to LLM providers.

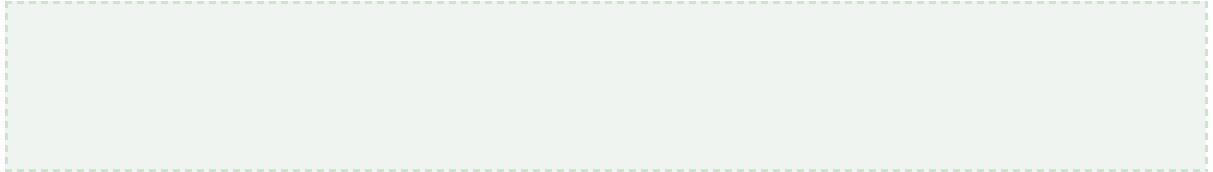
5. Feature List

The following 25 features constitute the complete scope of Buniyadi, grouped by functional area.

Core AI & Onboarding

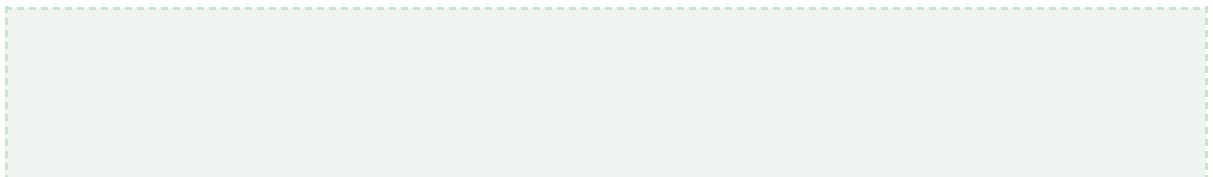
Feature 1: Conversational Idea Ingestion

A single plain-text input field where founders describe their startup idea in natural language. No forms, no dropdowns, no sign-up friction. The founder types and the system begins building.



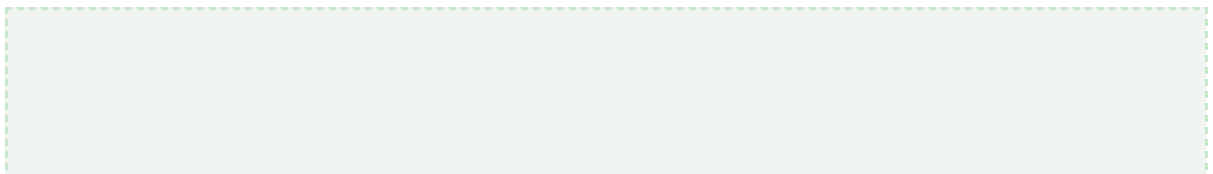
Feature 2: Progressive Context Rendering

As the FastAPI service processes the founder's prompt, business insight cards (industry, model, audience, risks) appear on screen one by one in real time. There is no full-page loading screen.



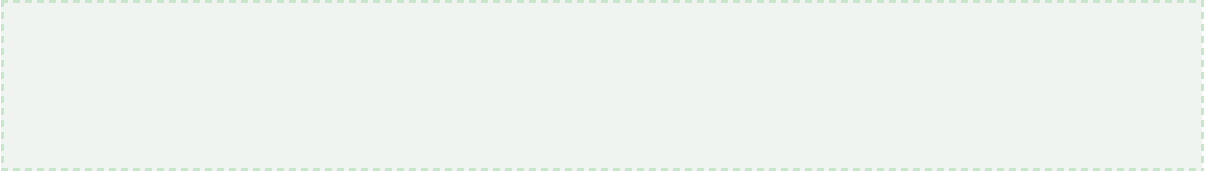
Feature 3: Context-Aware Global Memory

A centralized knowledge graph (using RAG) stores every detail of the startup. All subsequent AI interactions automatically reference this stored context, so the system always “remembers” prior decisions.



Feature 4: Knowledge Base Upload

Founders can upload external PDFs such as market research or user interview transcripts. The FastAPI service ingests these documents and incorporates them into the global knowledge base to further ground AI outputs.



The Ripple Effect (Dynamic Linking)

Feature 5: Global Variable Syncing

When a founder edits a core parameter — for example, changing the target audience from “Students” to “Professionals” — the system automatically recalculates and updates all dependent modules including pricing, marketing strategy, and brand tone.



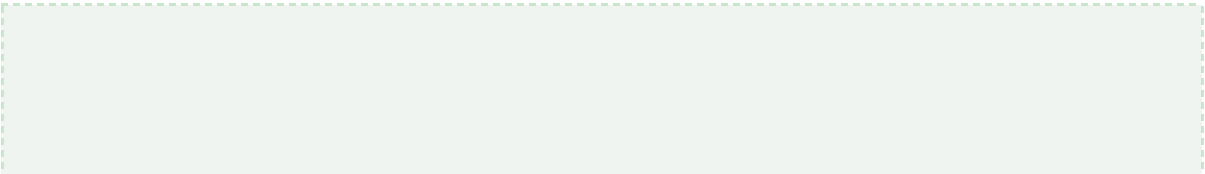
Feature 6: Explainable AI UI

Every AI-generated output (a colour palette, a pricing tier, a marketing channel) includes an accompanying tooltip or text block explaining the strategic reasoning behind it. Founders always understand why a suggestion was made.



Feature 7: Conflict Resolution Prompt

If a manual edit contradicts the overall strategy (e.g., a premium price point set for a budget-sensitive student audience), the system detects the inconsistency, flags it, and suggests corrective options.



Feature 8: Inline Magic Edit

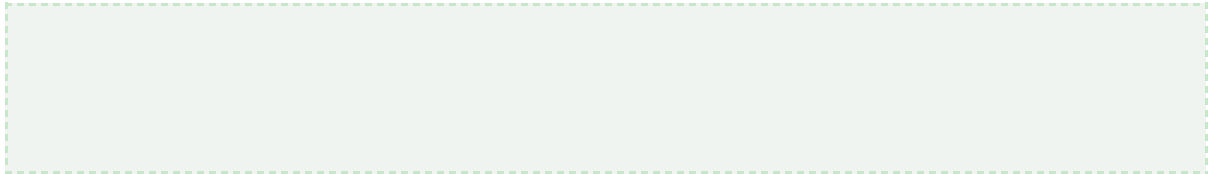
Users can highlight any specific text block within a room and prompt the AI to rewrite, expand, or pivot just that section, while the global context remains untouched and consistent.



Workspace Rooms (Modules)

Feature 9: Business Strategy Room

A dynamic canvas auto-generating the startup's Problem Statement, Proposed Solution, Unique Value Proposition, and Business Model based on the initial idea and stored knowledge base.



Feature 10: Competitor Intelligence Room

Auto-discovery of real-world competitors with a generated comparison matrix detailing their strengths, weaknesses, and pricing models, giving founders a clear picture of the competitive landscape.



Feature 11: Audience & Persona Room

Auto-generated, editable customer personas covering pain points, demographics, and buying behaviours. Founders can refine personas directly, triggering a Ripple Effect across dependent rooms.



Feature 12: Brand Identity Room

Generation of brand voice, typography recommendations, and colour palettes. Each choice is accompanied by an explanation of how it positions the startup against competitors in the identified market.



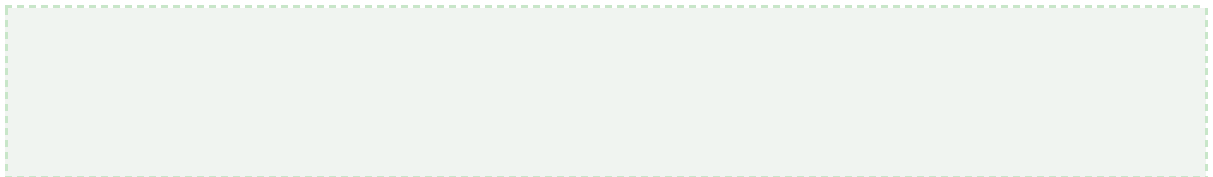
Feature 13: Marketing & Launch Room

Auto-generation of Go-To-Market (GTM) strategies, recommended acquisition channels, and actionable launch checklists tailored to the detected audience and business model.



Feature 14: Financials & Pricing Room

Suggested revenue models and pricing tiers derived from the detected target audience and competitor benchmarks, helping founders make data-informed monetisation decisions from day one.



Version Control & Tracking

Feature 15: Decision Timeline

A chronological version-control log tracking every pivot, manual edit, and AI generation. Founders can scroll back through time to see exactly how their idea has evolved since the first submission.



Feature 16: State Reversion

The ability to roll back the entire workspace, or individual rooms, to any previous state recorded on the Decision Timeline — functioning like a startup-specific version of Git.



Feature 17: Proactive Welcome Back Dashboard

A dynamic home screen that analyses changes since the founder's last login and proactively surfaces next-step suggestions or flags any new inconsistencies detected in the workspace.



Documents & Exporting

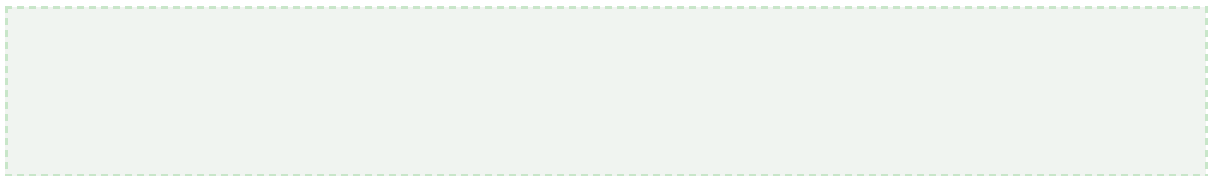
Feature 18: Living Business Documents

Auto-updating document templates (Executive Summary, Pitch Deck, Business Plan) that remain synchronized with the live workspace variables. Any change in the workspace is immediately reflected in the documents.



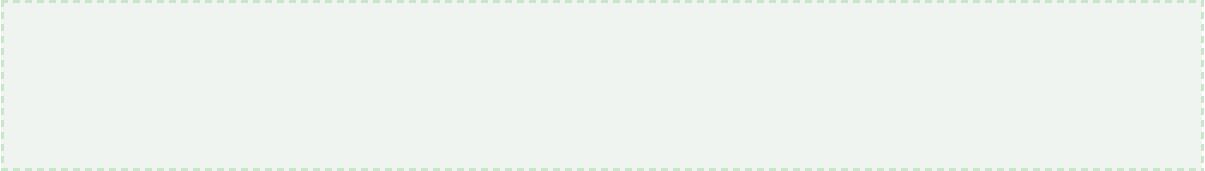
Feature 19: AI-Ready Context Export

Exports the startup's entire context as an optimized Markdown or JSON file, formatted to be used directly as a system prompt in external AI tools such as Cursor or ChatGPT.



Feature 20: Smart PDF Export

One-click generation of professional Investor Reports or Brand Guideline PDFs based on the current state of the workspace, ready to be shared with external parties.



Feature 21: Read-Only Investor Link

Generates a secure, live-updating URL that grants external viewers (mentors, investors) read-only access to specific workspace rooms such as the Executive Summary or Pitch Deck.



Collaboration & Infrastructure

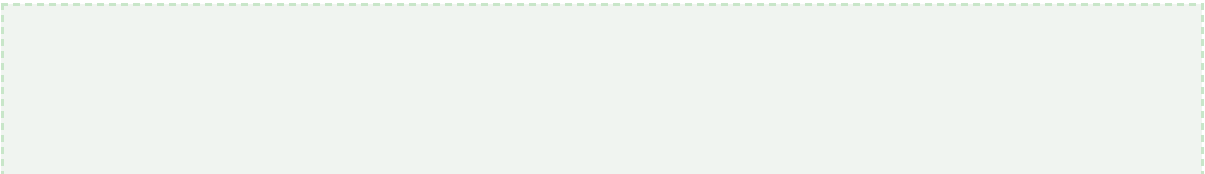
Feature 22: Multiplayer Collaboration

Real-time concurrent editing via WebSocket allows co-founders to work in the same workspace simultaneously, with changes visible to all participants instantly.



Feature 23: Role-Based Access Control (RBAC)

Permission settings allowing workspace owners to invite users as Co-Founders (full edit access), Mentors (comment-only access), or Investors (view-only access).



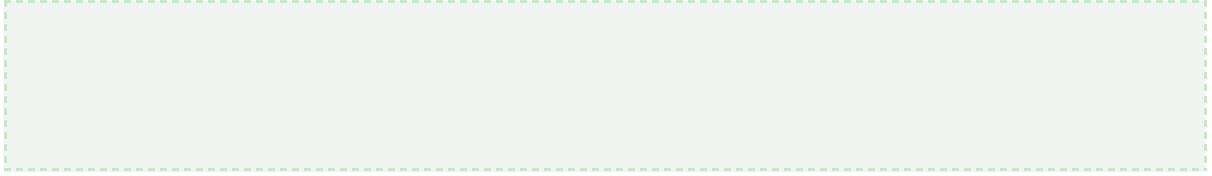
Feature 24: Unified Workspace Search

A global search bar enabling founders to instantly find specific decisions, competitor data, or document text across all rooms and all past versions in the Decision Timeline.



Feature 25: Task & Action Item Auto-Generator

The system reads the current state of the workspace and generates a dynamic, prioritized to-do checklist for the founding team, translating strategy into concrete, executable next actions.



6. Tentative Development Plan (Agile Methodology)

Sprint 1: Core Setup & Authentication

- Initialize GitHub repository and set up CI/CD pipelines.
- Configure Express server, MongoDB connection, and FastAPI microservice.
- Develop user registration, login, and JWT authentication flows.
- Establish the MVC project structure for both backends.

Sprint 2: AI Ingestion & Progressive UI

- Build the React landing page with the conversational text input.
- Develop FastAPI endpoints for LLM communication and initial idea breakdown.
- Implement progressive rendering of business model, audience, and risk cards on the frontend.
- Build the MongoDB workspace schema and wire up initial data storage.

Sprint 3: Workspace Rooms & Database Schemas

- Design and implement MongoDB schemas for nested workspace room data.
- Build the Business Strategy, Brand Identity, and Competitor Intelligence rooms.
- Build the Audience & Persona, Marketing & Launch, and Financials & Pricing rooms.
- Implement CRUD operations for manual inline editing within all rooms.

Sprint 4: Ripple Effect, Version Control & Document Generation

- Develop backend logic to detect variable changes and trigger global workspace recalculations via FastAPI.
- Implement the Conflict Resolution Prompt and Explainable AI tooltips.
- Build the Decision Timeline logging system, UI, and state reversion capability.
- Develop auto-updating document templates (Pitch Deck, Executive Summary, Business Plan).

Sprint 5: Export, Collaboration & Final Polish

- Implement PDF export and AI-ready Markdown/JSON context export.
- Build WebSocket-based real-time multiplayer collaboration.
- Set up RBAC and read-only investor link generation.
- Build the global workspace search, Welcome Back dashboard, and task auto-generator.
- Final QA, performance testing, and bug fixing.

7. Conclusion

This SRS provides a detailed outline for the design and development of “Buniyadi” — a comprehensive, AI-powered startup workspace built on a straightforward dual-backend architecture: the MERN stack for core application logic and FastAPI for all AI processing. By centralizing 25 features across business strategy, brand identity, competitor analysis, version control, document generation, and team collaboration into a single, continuously evolving platform, Buniyadi aims to become the indispensable single source of truth for every founder. Through the outlined five Agile sprints, the project will deliver a robust, scalable, and intelligent workspace that evolves alongside a founder’s startup journey.